The Backup and Clone Cards

The disk-image cards copy the node's SD card to a USB drive, restore an image to a drive, clone one drive to another, format a drive, and run backups on a schedule. This summary covers each card, the checks that stand before a job, the options, the image format, and how a job is watched and stopped.

BACKUP RESTORE CLONE FORMAT SCHEDULE

SUMMARY · READ-ONLY REVIEW

What is on the page

The disk-image section is one panel with a drive table and six cards. Each card has its own options, a command preview, and a start button. A single job console shows what the running job is doing.

CARD OR PART	WHAT IT DOES	MAIN ROUTE
Drives	Lists the disks, their partitions, mounts and filesystems. The boot disk is marked SYSTEM DRIVE.	`/api/diskimg/drives`
Backup	Copies the boot disk (the SD card) to an image file on a USB drive.	`/api/diskimg/backup/start`
Existing Backups	Lists the images on the drives, and downloads or deletes one.	`/api/diskimg/backup/list`, `/download`, `/delete`
Restore	Writes a chosen image from one drive onto another, spare drive.	`/api/diskimg/restore/start`
Clone	Copies this Pi's boot disk straight onto another USB drive.	`/api/diskimg/clone/start`
Full Format	Wipes a drive, makes a new partition table, and formats it.	`/api/diskimg/format/start`
Scheduled Backups	Runs a backup automatically at an interval, with retention.	`/api/diskimg/schedule`
Job	Shows the running job, its console, its progress, and cancel or reset.	`/api/diskimg/job`, `/job/cancel`, `/job/reset`
Mount, unmount, eject, preview	Manage a drive's mounts, power it off, and show the command before a job starts.	`/api/diskimg/mount`, `/unmount`, `/eject`, `/preview`

Import from USB Drive is a separate card, covered in Part 2 of this series. It also stages files, not disk images.

What stops a disk job from starting

- **The boot disk is never a target.** The server refuses it as the destination of a backup, restore, clone or format, and refuses to mount, unmount or eject it. The scheduled backup skips it. Backup and clone read from it and never write to it. If the boot disk cannot be worked out, backup, clone, restore and format refuse to start.
- **Only attached USB or removable drives.** Every destination must be a drive on the current list (a whole disk with a positive size, on a USB connection or flagged removable), and must not be write-protected. Format, clone and restore also need the drive to be unmounted first. Backup, restore, clone, format, mount, unmount and eject each need `confirm: true`, and enabling the schedule does too.
- **One job at a time.** A second job is refused (409) while one is running, or while node services are being restarted after a quiesced job. The server does not check for a running install at the start of a manual job (the scheduled backup does wait for installs).
- **Full Update and recording.** While a Full Update runs, every instmon POST request except an action is refused with 409. While Node Recorder is recording, the backup, format, clone, restore, mount, unmount, eject and schedule requests are refused with 409, and so are the editor save and the Confputter upload.
- **Tools present.** The job needs dd, lsblk, findmnt, mount, umount and partprobe. The dependency card lists what is missing and how to install it. The optional tools (for example ddrescue, rpi-clone, and the shrink tools) are needed only for those options.
- **A preview first.** The page shows the exact command a job would run, before it starts. The preview route builds it without running anything.

Copying the SD card to a USB drive

3.1 What is written

- The image is written to a folder named `instmon-backups` on the chosen drive. The name is `host name_date-time`, then `_label` if one is set, then `.img` (or `.img.enc` if encrypted). Before the copy, the drive must be at least as large as the copy and have that much free space, and stale `.part` files from earlier runs are cleared when no job is running.
- **How much is copied.** On a disk with a DOS (MBR) partition table, only the partitions are copied: up to the end of the last partition, rounded up to a whole MiB, and the destination needs to be at least that large. On a GPT disk, the whole disk is copied. The message says when only the partitions are copied.
- While copying, the file has a `.part` suffix. It is renamed to its final name only when the copy has finished, so an interrupted copy never looks like a finished backup.
- Next to each image is a sidecar file (`.json`). It records the format and version, the image name, its size, the hash algorithm and value, the source device and its size, the host name, the time, whether it is encrypted, whether it was shrunk, and the instmon version.
- A backup that is cancelled removes its partial file.

3.2 Options

OPTION	WHAT IT DOES
Label	A name for the image, used in its file name. Characters other than letters, digits, `_` and `-` become `_`, at most 40 characters.
Shrink	Makes the image smaller than the card. It works on the image file through a loop device, after the copy. The filesystem is checked with `e2fsck -f -y` first, and a filesystem that is still inconsistent is not shrunk. Then `resize2fs -M` shrinks it to its minimum, parted shrinks the partition to match, and the image file is truncated. It is skipped automatically when the disk layout does not allow it. It cannot be combined with encryption.
Verify	Takes a hash of the data as it is copied, then reads the image back from the drive and compares. The hash is SHA-256 (the default) or BLAKE2b. It cannot be combined with encryption.
Encrypt	Pipes the copy through openssl (`-aes-256-cbc -pbkdf2 -salt`). The passphrase is passed through an environment variable (`INSTMON_DISKIMG_PASS`), not on the command line. It cannot be combined with shrink, verify, the ddrescue backend or a throttle.
Pause node services (quiesce)	Stops the watchdog, the suite dashboards (SysMon keeps running), Asterisk and the bridges for the length of the copy, so fewer writes are in flight. The node is off the air until the job ends, and the services are then restarted. A check for package-manager locks runs first and stops the job if one is held. The image is crash-consistent, not a snapshot.
Freeze filesystem	A sub-option of the pause. It freezes the root filesystem (`fsfreeze -f /`) for the copy. It is available only with the pause. If the freeze fails, the job goes on without it.
Best effort	Keeps going past read errors (dd backend only). Unreadable sectors are padded with nulls.
Direct I/O and throttle	Direct I/O (clone with dd only) bypasses the cache. The throttle slows the copy; the backup card offers no throttle, 5, 10 or 20 MB a second. A throttle is not supported with encryption or with ddrescue.

Rescue backend	Uses ddrescue (`-d -b 4MiB`) instead of dd. It keeps a map file (`.ddrescue.map`). It cannot be combined with encryption or a throttle.

3.3 FAT drives

A FAT drive cannot hold a file larger than 4 GB. Before a copy to a FAT drive, the size is compared with 4 GB less one byte, and the job is refused with the reason if the image would not fit. The message says to format the drive as ext4 or exFAT on the Full Format card.

Writing an image back to a drive

1. **Pick the image.** It must be on a drive that is attached now. The card reads its sidecar and shows the size, the hash and whether it is encrypted. If the sidecar's size does not equal the file's size, the restore is refused as a damaged or incomplete backup.
2. **Pick the drive.** The boot disk is refused ("restore always targets a different, spare drive"). The drive that holds the image is refused too. The drive must be attached, not write-protected, unmounted, and at least as large as the image.
3. **Write.** The image is copied to the drive. An encrypted image needs the passphrase, and is decrypted on the way. For an encrypted image, verify and a throttle are not supported.
4. **Check the hash** when the sidecar has one. The image is checked against its saved checksum as it is read. A written image whose hash does not match is reported as a failure.
5. **Expand to fill** (optional). After the restore, the last partition is grown with parted to 100 % of the drive, the filesystem is checked with `e2fsck -f -y`, and it is grown with resize2fs.
6. **New disk IDs** (optional). The disk's identifiers are changed, so two cards restored from the same image do not share them.

A restore writes over the drive it is given, so anything on that drive is replaced.

Copying one drive to another

A clone copies this Pi's boot disk to another USB drive with no image file in between. The boot disk is always the source, and the destination must be attached, not write-protected, unmounted, and (for dd and ddrescue) at least as large as the copy (the partitions on a DOS disk, otherwise the whole disk). Three backends are offered:

BACKEND	WHAT IT DOES	VERIFY AND FREEZE
dd	A direct block copy. It supports best effort, direct I/O, a throttle, verify, expand and new IDs.	Both available
ddrescue	A rescue copy with a map file. No throttle.	As for dd, apart from best effort and direct I/O, which do not apply
rpi-clone	The rpi-clone tool, which syncs the Raspberry Pi system with its own options. It must be installed first. A new host name for the copy may be given (letters, digits and dashes, up to 63). It prepares the boot files first and refuses with 409 if that fails. No percentage is shown, only the console.	Verify, best effort, direct I/O and the throttle are turned off. Freeze is refused with 400

Expand and new IDs apply only to the dd and ddrescue clones. The size of the copy is known, so the progress can be shown. The pause of node services and the freeze work as in a backup.

Wiping and formatting a drive

- **Filesystems:** ext4 (mkfs.ext4), FAT32 (mkfs.vfat), or exfat (mkfs.exfat).
- **Before it starts:** the drive must be attached, not write-protected and unmounted, and the format needs `confirm: true`.
- **Quick or full:** a quick format writes zeros over the first 16 MB of the drive and runs `wipefs -a`. It is not a secure erase. A full format writes zeros over the whole drive in 4 MB blocks, so it takes much longer.
- **Steps:** after the wipe, parted writes a GPT label with one primary partition spanning the drive, then the filesystem tool formats it. The output is shown in the job console. If a needed tool is missing, the job stops with the tool's name.
- **The boot disk** is refused (Part 2).

A format destroys every file on the drive.

Backups that run on their own

7.1 Settings

SETTING	DEFAULT	MEANING
Enabled	Off	The schedule runs only when turned on.
Interval	24 hours	Time between backups. A backup is due when this much time has passed since the last run.
Destination	None	The drive. Its filesystem UUID is remembered, so a drive that changes its device name is still found.
Retention	5	How many scheduled images are kept. The oldest are deleted when the count is passed.
Shrink, verify, quiesce	Off	The same options as a manual backup.
Label	auto	The label in the file name.

7.2 When it waits

The scheduler checks every minute. A backup that is due does not start, and the reason is logged and kept, when:

- the node booted less than 15 minutes ago (so the network, tunnel and radio services can start first);
- a disk job is already running;
- a Full Update is running, a recording is in progress, or an install is running;
- no boot disk can be determined, no destination is set, or the destination drive is not attached.

The last skip and the last run (with its status and message) are kept in the schedule settings and shown on the card.

7.3 Retention

Only the scheduled images are pruned. They are recognised by their name ending in `.sched.img`. Manual backups are never pruned. The oldest scheduled images, by modification time, are removed beyond the retention count, along with their sidecars.

Watching a job

- **The console** keeps the last 300 lines of the job's output. The progress line of dd or ddrescue is turned into bytes done, rate and percentage.
- **The job record** is saved as the job runs, so when instmon restarts, it can tell the page what was running and how it ended (the last-job recovery).
- **Cancel** stops the copy by stopping its systemd unit. A cancelled backup deletes its partial file. A cancelled clone, restore or format leaves the destination partial. Cancel is refused while services are being stopped or restarted, while the filesystem is being thawed, and while the job is shrinking, formatting, expanding or changing disk IDs. Cancel during a verify leaves the copy as it is.
- **Reset** clears a finished job's record from the page.
- **Eject** syncs the drive, unmounts everything on it, and powers the USB device off. It needs `confirm: true`, and is refused for the boot disk and while a job is busy. If the power-off fails after the unmount, the answer is a warning that it is safe to remove.

Copies run as systemd transient units, so they keep going if the instmon process is restarted.

What the disk cards show

9.1 The boot disk is protected in code

The checks refuse the boot disk as a target in restore, clone, format and eject. The book can say the card cannot overwrite the disk the node is running from.

9.2 An image is only trusted when its hash matches

A backup is written with a `.part` name and renamed only when it finishes, and its hash is recorded in the sidecar. A restore that checks the hash can show that the image was not damaged on the drive.

9.3 Scheduled backups wait for the node to settle

The 15-minute boot wait and the skip rules keep a scheduled copy from competing with a Full Update, a recording or an install. The book should say that a backup can be skipped, and where the reason is shown.

9.4 Not checked

- The dd, ddrescue and rpi-clone command lines were read in part, and were not run.
- The partition and identity changes on a real card, and the resize results.
- The exact progress parsing for each backend.
- The restore card's warnings and the size checks in the browser.